Bihar Engineering University, Patna

B.Tech 4th Semester Examination, 2024

Course: B.Tech
Code: 101402

Subject: Disaster Preparedness & Planning

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 *Choose the correct answer of the following (any seven question only):* **[2 x 7 = 14]**

(a) What is the correct definition of a disaster?
- (i) A sudden event causing significant damage or loss
- (ii) A minor inconvenience in daily life
- (iii) A naturally occurring phenomenon without any impact
- (iv) A situation that does not require emergency response

(b) Which of the following is NOT a type of hazard?
- (i) Natural
- (ii) Technological
- (iii) Political
- (iv) Biological

(c) Which factor determines the severity of a disaster?
- (i) The frequency of similar events
- (ii) The level of preparedness and response
- (iii) The type of media coverage
- (iv) The number of early warning signs

(d) The difference between hazard and disaster is:
- (i) Hazards are potential threats, while disasters cause actual damage
- (ii) Disasters are always natural, while hazards are man-made
- (iii) Hazards cannot be prevented, while disasters can
- (iv) Disasters occur only in developing countries

(e) The frequency of a disaster event refers to:
- (i) The number of people affected by it
- (ii) How often a particular hazard occurs in a given area
- (iii) The severity of the hazard
- (iv) The speed of emergency response

(f) What is the primary goal of disaster prevention?
- (i) To completely eliminate all hazards
- (ii) To reduce risks by addressing root causes
- (iii) To focus on emergency response only
- (iv) To provide relief supplies after a disaster

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(g) Which of the following is an example of a natural hazard?
- (i) Nuclear explosion
- (ii) Earthquake
- (iii) Oil spill
- (iv) Industrial accident

(h) Which term refers to measures taken before a disaster to reduce its impact?
- (i) Mitigation
- (ii) Response
- (iii) Recovery
- (iv) Rehabilitation

(i) Which of the following is an example of a social vulnerability factor?
- (i) Weak building structures
- (ii) Poor emergency planning
- (iii) Low income and lack of education
- (iv) Geographic location near a fault line

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(j) Which of the following describes an effective disaster preparedness strategy?
(i) Investing in early warning systems
(ii) Educating communities on emergency response
(iii) Conducting disaster drills and training
(iv) All of the above

Q.2 (a) Define disasters and explain their classification. How do natural and man-made disasters differ in terms of impact and mitigation strategies? [7]

(b) Explain the different types of earthquakes and their impact. How does India's geographical position make it vulnerable to seismic activity? [7]

Q.3 (a) Explain the concept of hazard and vulnerability. How do mountain and coastal areas in India differ in terms of disaster vulnerability? [7]

(b) What are the key challenges in disaster management in India? Suggest measures to improve preparedness and response strategies for both natural and man-made disasters. [7]

Q.4 (a) How do disasters affect human health? Explain the short-term and long-term health impacts of disasters, including psycho-social issues. [7]

(b) Discuss the role of hazard location in disaster vulnerability. How do geographical and environmental factors influence the severity and frequency of disasters? [7]

Q.5 (a) Explain the link between climate change and disasters. How does climate change contribute to the increasing frequency and intensity of urban disasters? [7]

(b) How can governments and communities improve disaster resilience? Suggest strategies for reducing disaster impacts at global, national, and local levels. [7]

Q.6 (a) Differentiate between structural and non-structural measures in disaster risk reduction. Provide examples of each and explain their significance in disaster management. [4]

(b) Describe the role of early warning systems in disaster risk reduction. How do technological advancements enhance early warning and disaster preparedness? [10]

Q.7 Explain the roles and responsibilities of various stakeholders in disaster management, including the government, community, local institutions, NGOs, and international organizations. [14]

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Q.8 Analyze the relationship between disasters, environment, and development. How can improper development planning increase disaster risks, and what measures can be taken to balance development with environmental sustainability? [14]

Q.9 Write short notes on ***any two*** of the following: - [7 x 2 = 14]
(a) Environmental Sustainability
(b) Post-disaster Management
(c) Disaster Vulnerability
(d) Disaster Risk Reduction

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